

Smart Food Safety & Hygiene Monitoring System

Technical Proposal

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Executive Summary

This project addresses the lack of continuous monitoring in restaurants, which often leads to hygiene violations and foodborne illnesses. The proposed solution is an AI-powered system that automatically detects hygiene issues and provides real-time insights to ensure compliance before health inspections.

Project Understanding

Restaurants typically follow hygiene guidelines only when inspections are due. Our system will act as a constant monitoring tool, ensuring standards are maintained at all times, thereby protecting consumer health and reducing risk for business owners.

Objectives

- Develop an AI model to detect hygiene issues from images/sensor data.
- Build a real-time dashboard for monitoring and alerts.
- Enable early detection to prevent costly violations.
- Deliver a scalable and reliable solution suitable for restaurants of different sizes.

Proposed Solution

- Data Collection: Capture images/sensor data from kitchens and storage areas.
- Preprocessing: Image cleaning, augmentation, and labeling.
- AI/ML Model: Train computer vision models to detect unclean surfaces, improper storage, and staff hygiene violations.
- Dashboard & Alerts: Interactive dashboard using Dash/React/Flask to show hygiene status and generate alerts.
- Testing & Validation: Simulate deployment in test kitchens to refine performance.

System Architecture

The system consists of three layers: (1) Data Layer for collecting and preprocessing inputs, (2) Intelligence Layer where AI models analyze data, and (3) Presentation Layer that delivers results through a dashboard with alerts and reports.

Tools & Technologies

- Programming Languages: Python, JavaScript
- Frameworks: TensorFlow/PyTorch, Flask/Dash, React
- Databases: PostgreSQL (optional for logging)
- Version Control: Git/GitHub

Deliverables

- Preprocessed dataset
- AI model for hygiene detection
- Web-based dashboard with alerts
- Technical documentation and user guide

Timeline & Milestones

- Weeks 1–2: Data collection & preprocessing
- Weeks 3–4: AI model development
- Week 5: Dashboard development & integration
- Week 6: Testing, validation, and final reporting

Risks & Mitigation

- Insufficient data quality → Use augmentation and synthetic data.
- Model accuracy issues → Continuous retraining and validation.
- System integration delays → Agile approach with weekly milestones.

Success Criteria

- Model achieves at least 85% accuracy in detecting hygiene issues.
- Dashboard displays real-time alerts with less than 5s latency.
- System validated in at least one simulated kitchen environment.

Expertise

This project leverages expertise in machine learning, computer vision, and web development. With a strong background in data preprocessing and visualization, I am well-prepared to design and deliver a practical, impactful solution for food safety monitoring.